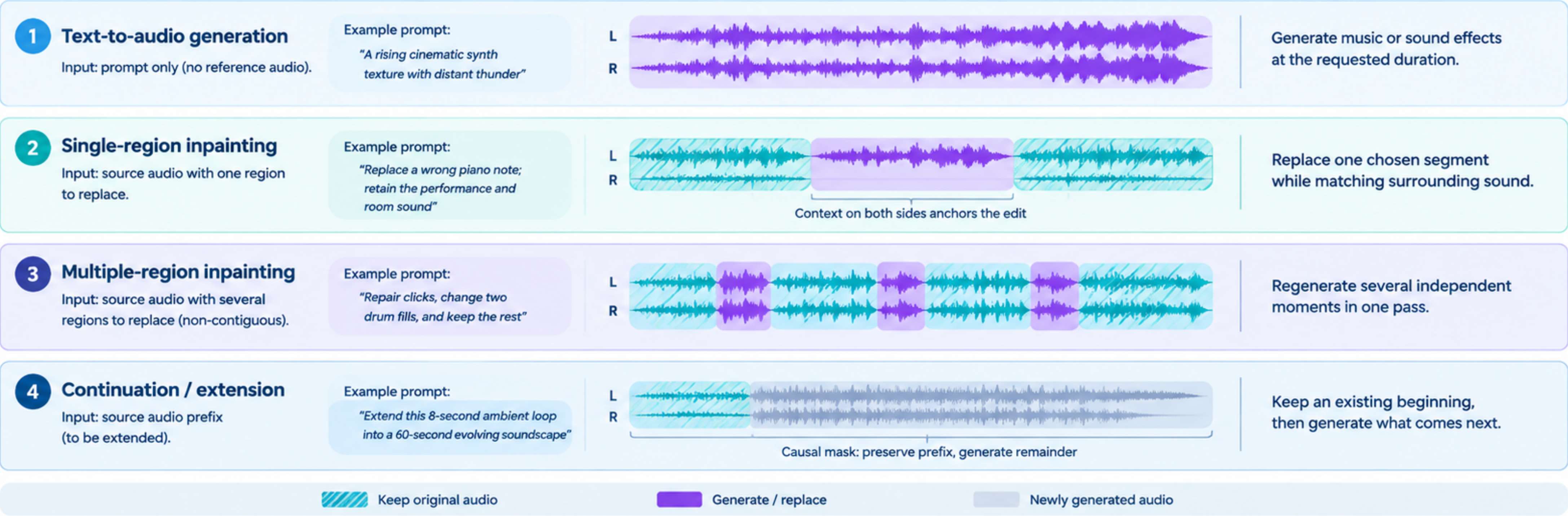
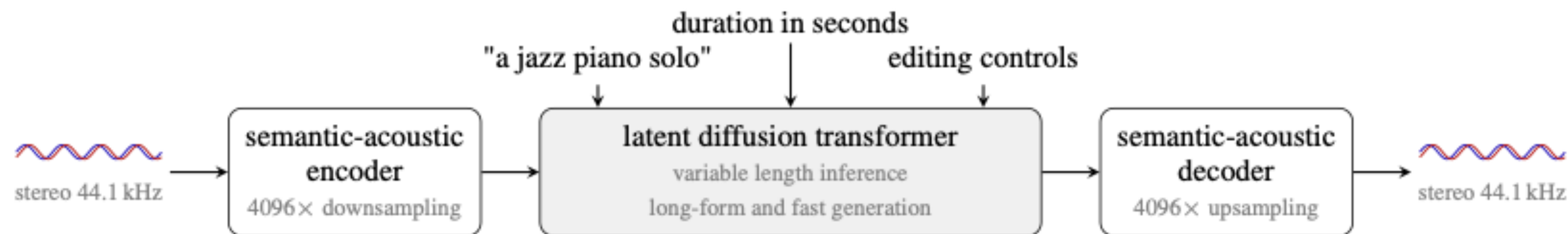


Stable Audio 3

Stable Audio 3 - Musical Application Overview



Stable Audio 3 - Architecture Overview

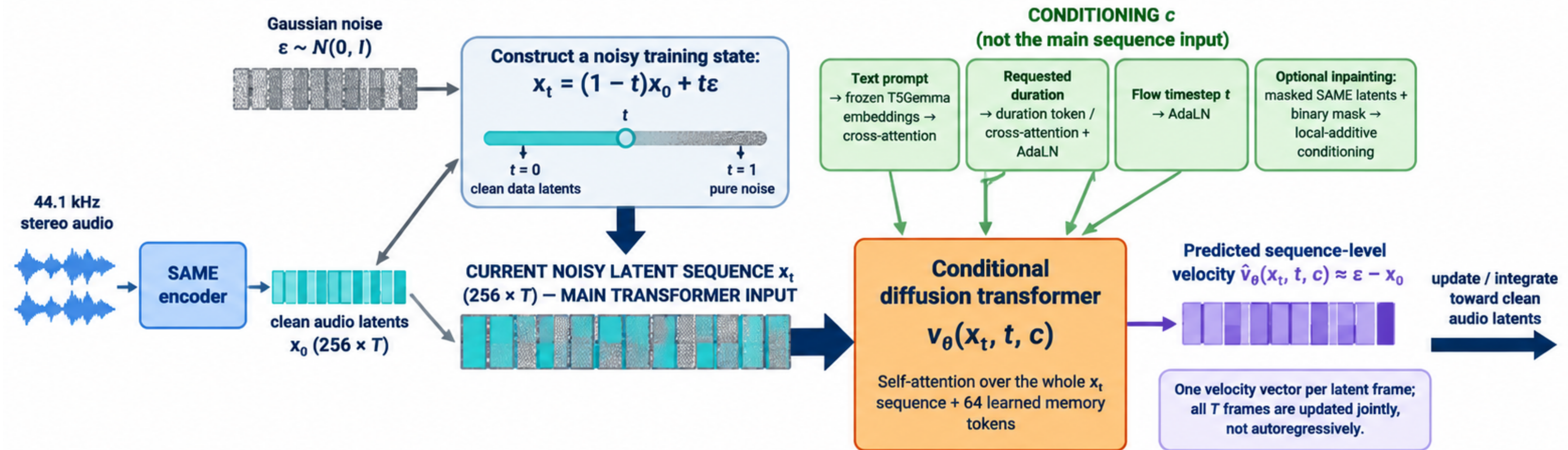


Stable Audio 3 text-to-audio with variable-length generation and editing via inpainting.

	maximum length	# learnable parameters	H200 inference time	open weights	licensed data	generation music	sfx
small-music	2m	459M	0.44s	✓	✓	✓	—
small-sfx	2m	459M	0.44s	✓	✓	—	✓
medium	6m 20s	1.4B	1.31s	✓	✓	✓	✓
large	6m 20s	2.7B	1.80s	—	✓	✓	✓

<https://arxiv.org/pdf/2605.17991>

Stable Audio 3 - Operation Overview

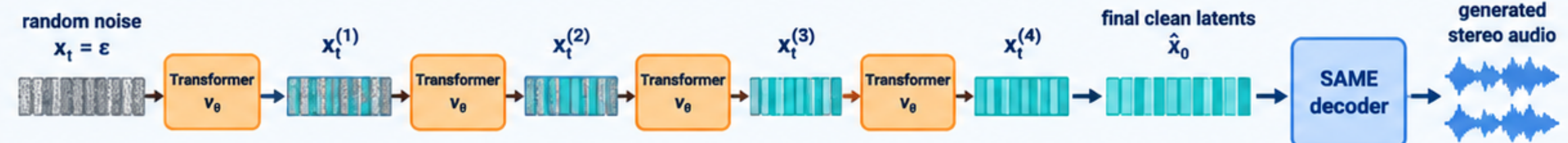


Training

$$\text{MSE}(\hat{v}_\theta, \epsilon - x_0)$$

Learn a conditional vector field from noise toward valid audio latents.

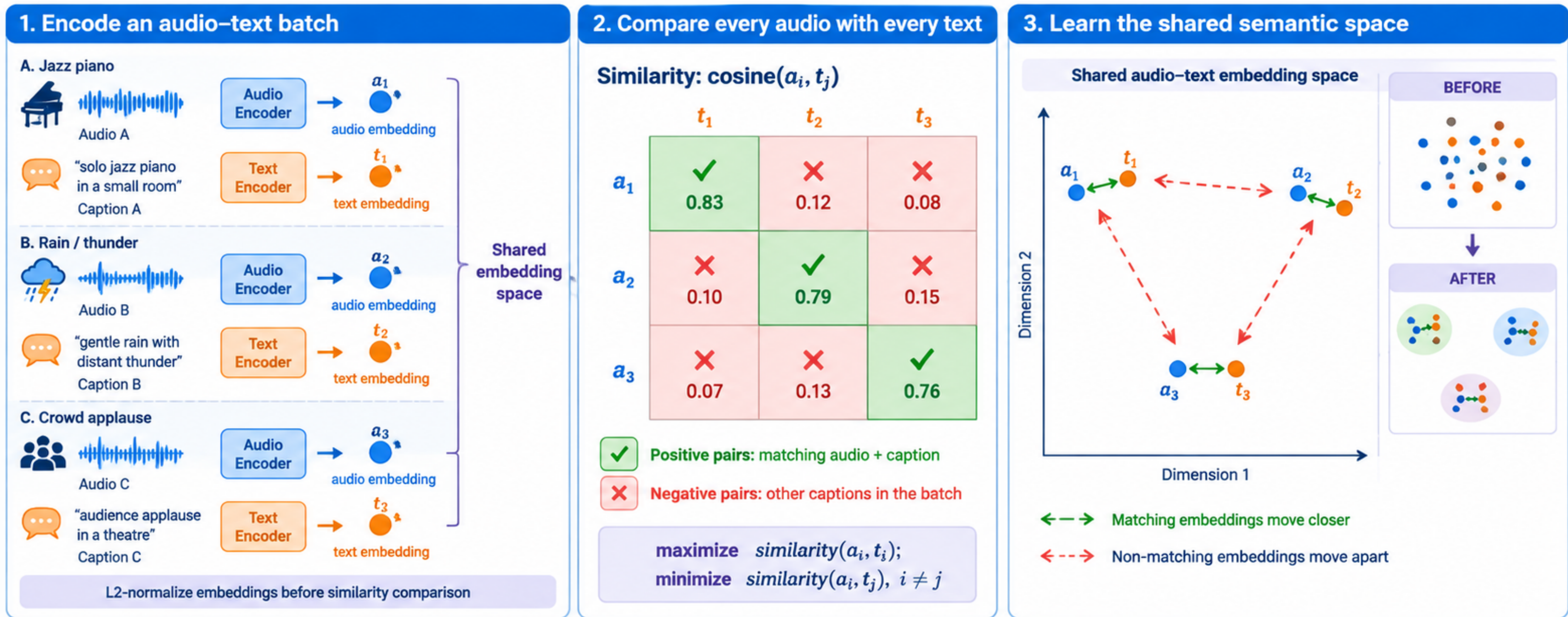
Generation



Post-trained Stable Audio 3: typically 8-step ping-pong sampling.

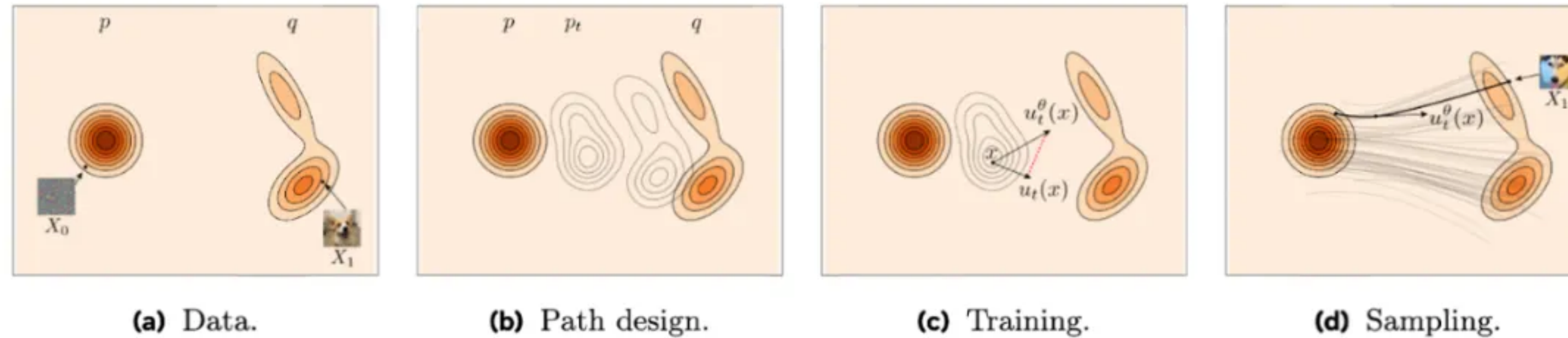


Stable Audio 3 - SAME - Contrastive Audio Text Pretraining

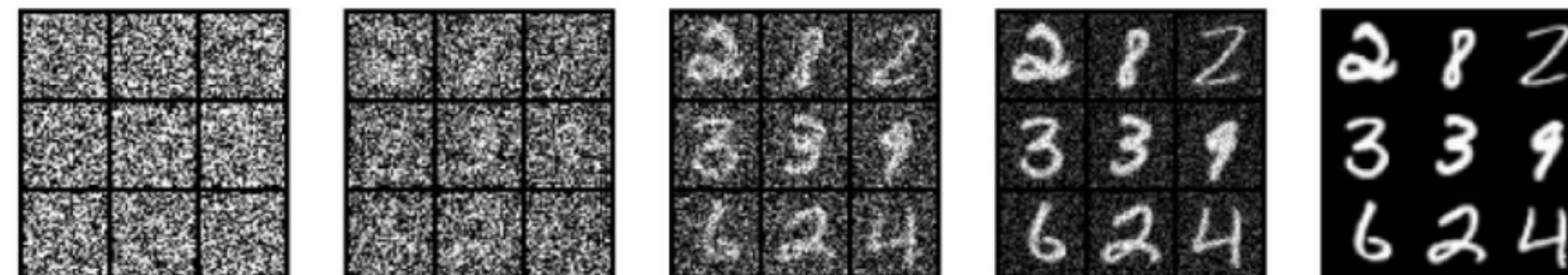
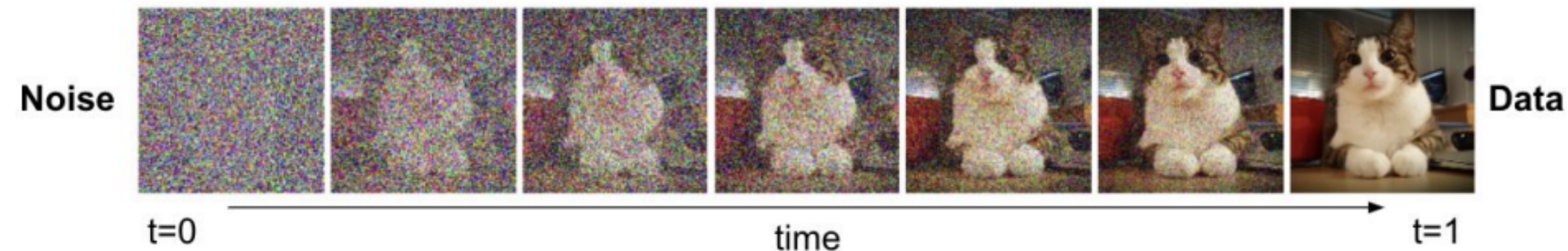


Stable Audio 3 - Flow Matching Principle

$$\dot{x} = f_{\theta}(x, t)$$

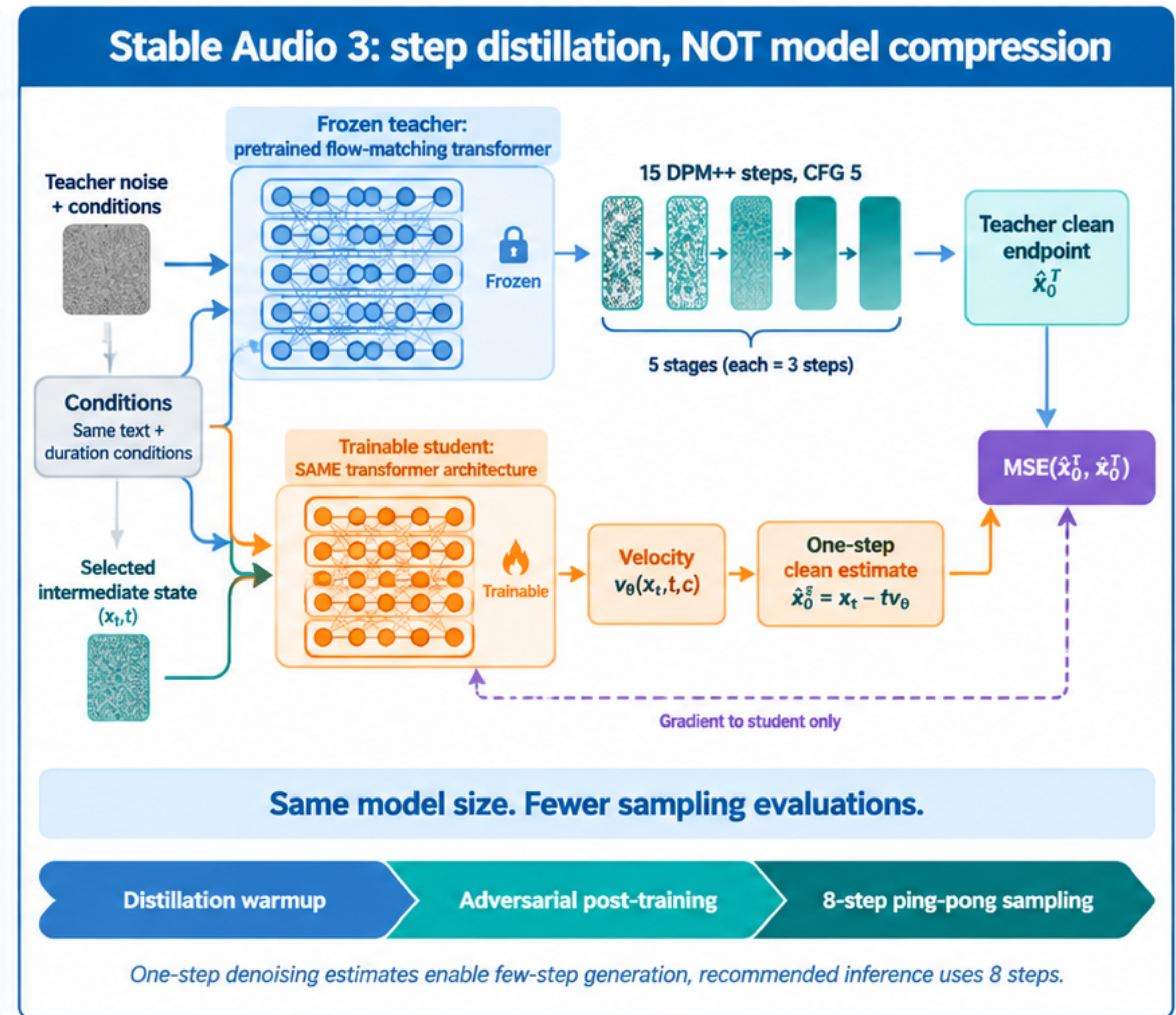
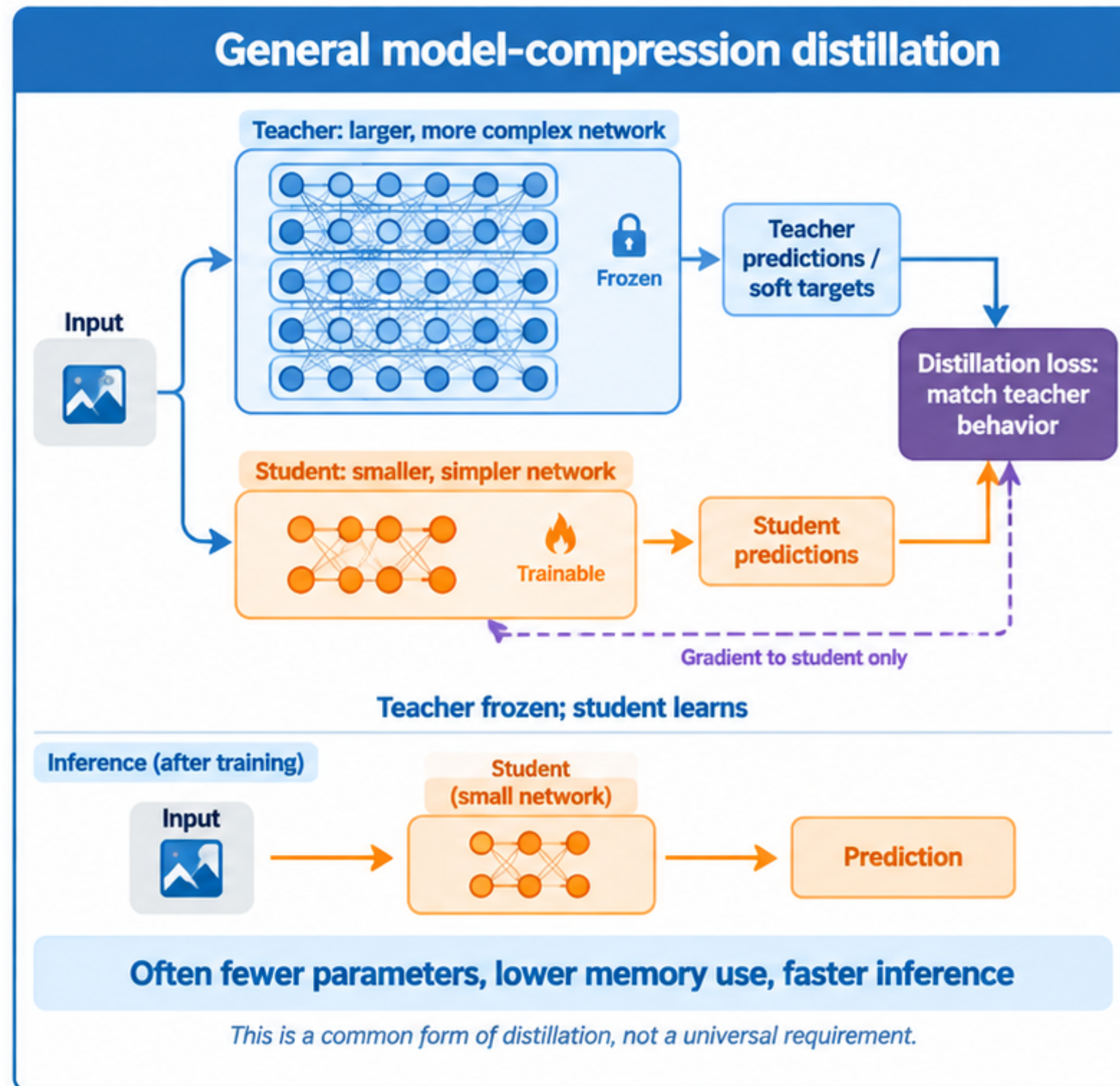


Convert a Source Distribution into a Target Distribution
From: Lipman et al 2024: Flow Matching Guide and Code

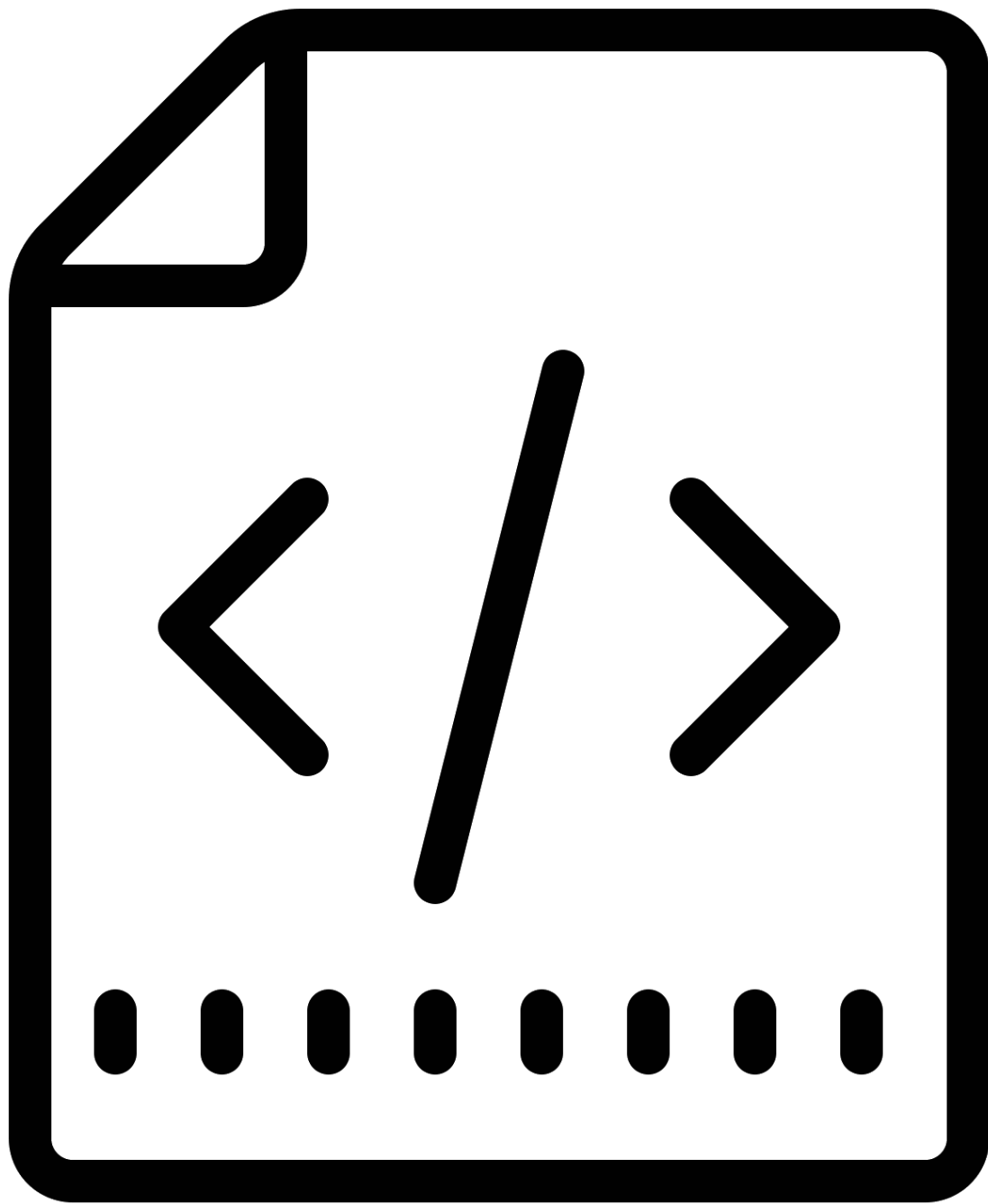


The Path from Noise to Data
From: Holderrieth and Erives, MIT IAP 2025

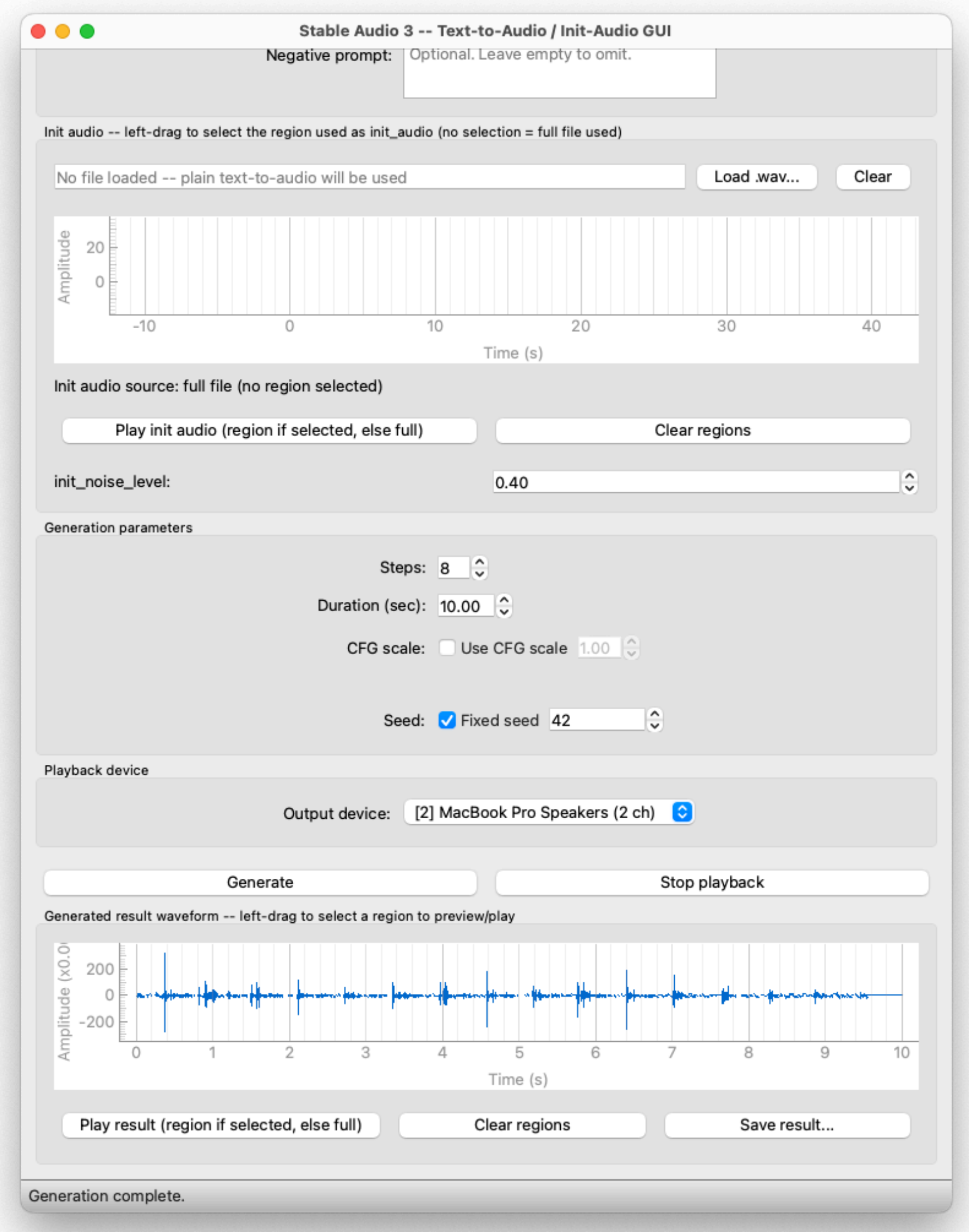
Stable Audio 3 - Knowledge Distillation



Stable Audio 3 - Demos

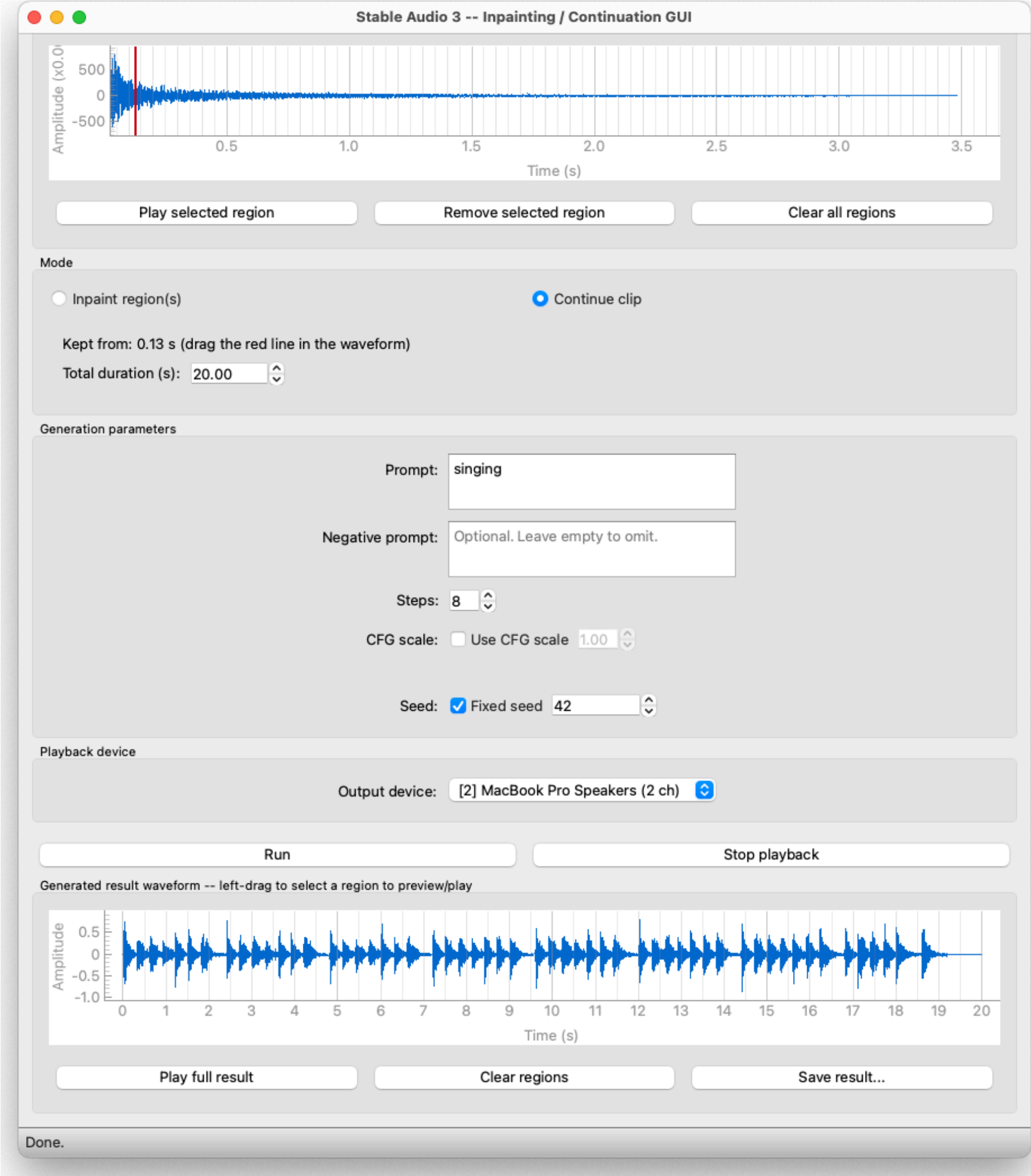


AI-Sound / StableAudio3 /
stable_audio_3_hello_world.py

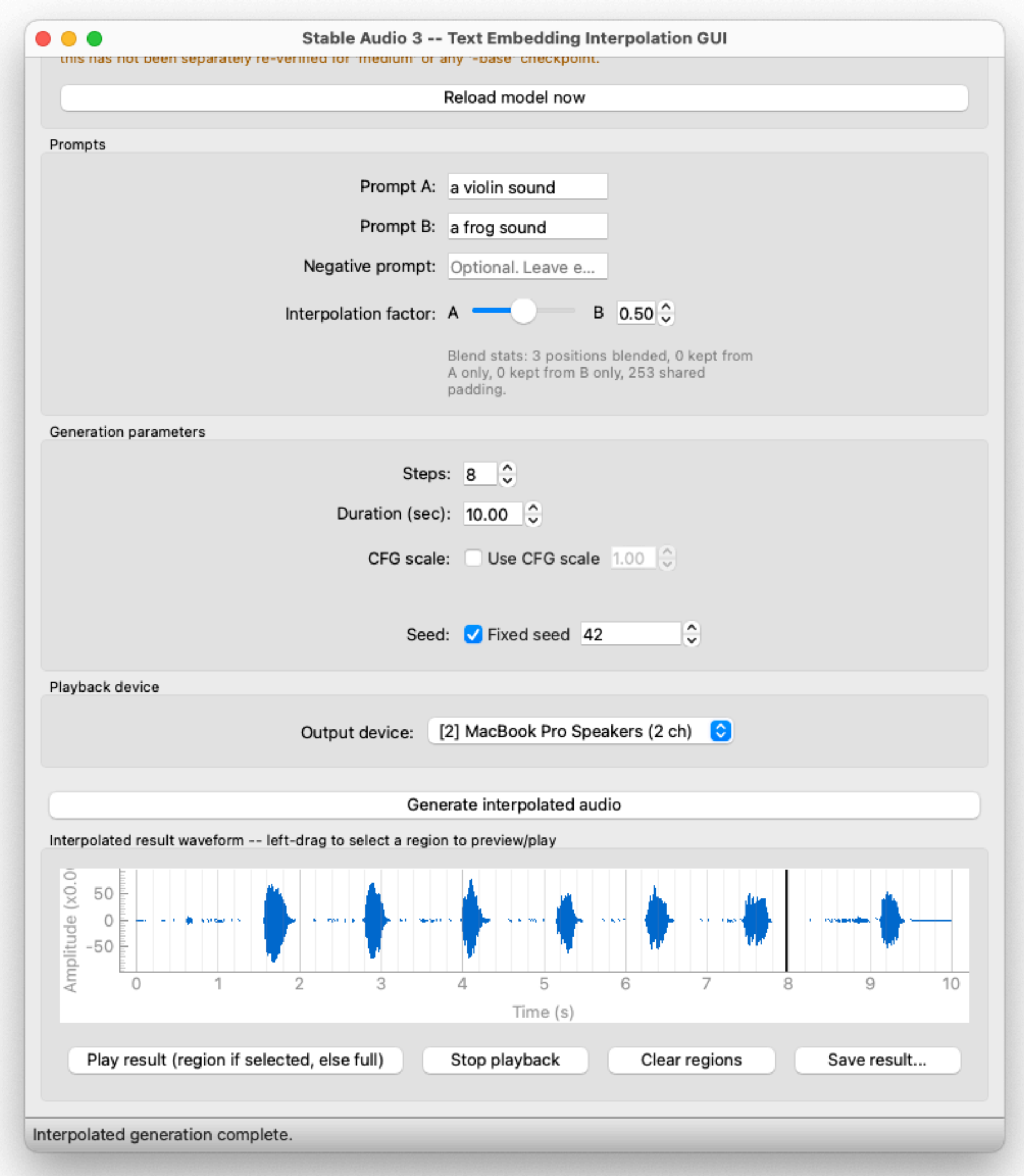


AI-Sound / StableAudio3 /
stable_audio_3_gui_text_to_audio.py

Stable Audio 3 - Demos



AI-Sound / StableAudio3 /
stable_audio_3_gui_inpaint_continue.py



AI-Sound / StableAudio3 /
stable_audio_3_gui_interpolation.py